

MATH 3115 MSc & SED COMPLEX ANALYSIS

Lectures 8 & 9: ANALYTIC FUNCTIONS

Course Lecturer: M. KANYIKA

Applied Studies Department- MUST

25th May, 2025

Lesson Outcome

By the end of this lesson you should be able to:

- Define Fundamental theorem of Algebra .
- Describe Cauchy-Riemann equations
- Describe Analytic function
- Solve problems involving Harmonic functions
- Define Function of complex variable.
- solve problems related to function of complex numbers
- Describe limits of function of complex variables
- Describe continuous function of complex variables

Fundamental theorem of Algebra

Fundamental theorem of Algebra

The fundamental theorem of Algebra states that every non-constant polynomial equation with complex coefficients has at least one complex root.

Moreover, if the degree of the polynomial is n , it will have exactly n complex roots, counting multiplicities.

Example 1

Consider the quadratic equation $x^2 + 2x + 5 = 0$. Find the roots of the polynomials.

Fundamental theorem of Algebra

Solution to example 1

The quadratic formula to solve for x is:

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Since $a = 1$, $b = 2$, and $c = 5$ then into the formula we have

$$x = \frac{-2 \pm \sqrt{2^2 - 4(1)(5)}}{2(1)}$$

Simplifying the expression inside the square root:

$$x = \frac{-2 \pm \sqrt{4 - 20}}{2}$$

$$x = \frac{-2 \pm \sqrt{-16}}{2}$$

Fundamental theorem of Algebra

Solution to Example 1

Since we have a negative number under the square root, we get an imaginary r

$$\sqrt{-16} = 4i$$

Now substitute this back into the equation:

$$x = \frac{-2 \pm 4i}{2}$$

Now, divide each term by 2 :

$$x = \frac{-2}{2} \pm \frac{4i}{2}$$
$$x = -1 \pm 2i$$

The two roots are:

$$x = -1 + 2i \quad \text{and} \quad x = -1 - 2i$$

Example 2

Consider the cubic equation: $f(z) = z^3 - 1 = 0$. Find the roots of the polynomial

Solution to example 2

Factoring $z^3 - 1$ using the difference of cubes formula we get:

$$z^3 - 1 = (z - 1)(z^2 + z + 1)$$

So the equation becomes:

$$(z - 1)(z^2 + z + 1) = 0$$

Fundamental theorem of Algebra

Solution to example 2 CONT'D

1. For $z - 1 = 0$, we have $z = 1$. Thus our first root.
2. For $z^2 + z + 1 = 0$: We now solve the quadratic equation $z^2 + z + 1 = 0$ using the quadratic formula: $z = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$
For $a = 1, b = 1$, and $c = 1$, we have:

$$\begin{aligned} z &= \frac{-1 \pm \sqrt{1^2 - 4(1)(1)}}{2(1)} \\ &= \frac{-1 \pm \sqrt{1 - 4}}{2} \\ &= \frac{-1 \pm \sqrt{-3}}{2} \end{aligned}$$

Since $\sqrt{-3} = i\sqrt{3}$, Then we have:

$$z = \frac{-1 + i\sqrt{3}}{2} \quad \text{and} \quad z = \frac{-1 - i\sqrt{3}}{2}$$

Fundamental theorem of Algebra

Solution to example 2 CONT'D

Therefore, the three roots of the equation $z^3 - 1 = 0$ are:

$$z = 1, \quad z = \frac{-1 + i\sqrt{3}}{2}, \quad z = \frac{-1 - i\sqrt{3}}{2}$$

ANALYTIC FUNCTIONS

Cauchy-Riemann equations definition

Cauchy-Riemann equations are a set of conditions that a function must satisfy in order to be complex differentiable (holomorphic) at a point.

Cauchy-Riemann equations CONT'D

If a complex function $f(z)$ is written as:

$$f(z) = u(x, y) + iv(x, y)$$

where:

- $u(x, y)$ is the real part of the function,
- $v(x, y)$ is the imaginary part of the function,
- and $z = x + iy$, where x and y are real numbers (the real and imaginary parts of z), then
- The Cauchy-Riemann equations say that for the function to be differentiable (holomorphic) at a point, the following must be true:

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} \quad \text{and} \quad \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$$

Example

Consider the complex function: $f(z) = z^2 = (x + iy)^2$.
Verify whether $f(z)$ satisfy the Cauchy-Riemann equations

Solution

- $f(z) = (x + iy)^2 = x^2 - y^2 + 2ixy$.
- Real part $u(x, y) = x^2 - y^2$
- and the imaginary part $v(x, y) = 2xy$.
- By Cauchy-Riemann equations
- $\frac{\partial u}{\partial x} = 2x$, and $\frac{\partial v}{\partial y} = 2x$,
- $\frac{\partial u}{\partial y} = -2y$, and $\frac{\partial v}{\partial x} = 2y$.
- Since the Cauchy-Riemann equations are satisfied then $f(z) = z^2$ is holomorphic

Analytic function definition

A function $f(z)$ is analytic at a point z_0 if there exists a neighborhood around z_0 such that $f(z)$ is holomorphic (complex differentiable) at all points in this neighborhood.

Example

Consider the complex function:

$$f(z) = z^2 = x^2 - y^2 + 2ixy$$

Verify whether $f(z)$ is analytic in a complex plane.

solution

$$f(z) = (x + iy)^2 = x^2 - y^2 + 2ixy$$

- The real part $u(x, y) = x^2 - y^2$
- and the imaginary part $v(x, y) = 2xy$
- Since the Cauchy-Riemann equations for $u(x, y)$ and $v(x, y)$ are: 1.

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} \quad 2. \quad \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$$

Compute the partial derivatives of $u(x, y) = x^2 - y^2$ and $v(x, y) = 2xy$:

$$\frac{\partial u}{\partial x} = 2x, \quad \frac{\partial u}{\partial y} = -2y$$

$$\frac{\partial v}{\partial x} = 2y, \quad \frac{\partial v}{\partial y} = 2x$$

1. $\frac{\partial u}{\partial x} = 2x$ and $\frac{\partial v}{\partial y} = 2x$, so the first equation is satisfied.
2. $\frac{\partial u}{\partial y} = -2y$ and $-\frac{\partial v}{\partial x} = -2y$, so the second equation is satisfied.

Since both Cauchy-Riemann equations hold, the function $f(z) = z^2$ is analytic.

Definition 1

A function is harmonic if both its real and imaginary parts satisfy the Laplace equation:

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$$

$$\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} = 0$$

Example

Consider the complex function

$f(z) = z^2 = x^2 - y^2 + 2ixy$. Determine whether the real part and imaginary of $f(z)$ is harmonic.

Solution

For $u(x, y) = x^2 - y^2$:

First, Lets compute the second partial derivatives of $u(x, y)$:

$$\frac{\partial^2 u}{\partial x^2} = 2, \quad \frac{\partial^2 u}{\partial y^2} = -2$$

Checking the Laplace equation for $u(x, y)$:

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 2 + (-2) = 0$$

Since the Laplace equation is satisfied, $u(x, y) = x^2 - y^2$ is harmonic.

For $v(x, y) = 2xy$: Now, compute the second partial derivatives of $v(x, y)$:

$$\frac{\partial^2 v}{\partial x^2} = 0, \quad \frac{\partial^2 v}{\partial y^2} = 0$$

Solution CONT'D

Check the Laplace equation for $v(x, y)$:

$$\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} = 0 + 0 = 0$$

$$\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} = 0$$

Since the Laplace equation is satisfied, $v(x, y) = 2xy$ is also harmonic.

FUNCTION OF COMPLEX VARIABLE

Definition

A function f of the complex variable z is a rule that assigns to each value z in a set D one and only one complex value W . We write

$$w = f(z)$$

where w the image of z under f .

- The set D is called the domain of definition of f ,
- The set of all images $R = \{w = f(z) : z \in D\}$ is called the range of f .

Function of Complex numbers

Complex value function

- Just as z can be expressed by its real and imaginary parts, $z = x + iy$.
- We write $w = u + iv$, where u and v are the real and imaginary parts of w , respectively.
- This gives us the representation $f(x + iy) = u + iv$.
- Since u and v depend on x and y , they can be considered to be real functions of the real variables x and y ; that is: $u = u(x, y)$ and $v = v(x, y)$.
- In short, a complex value function $W = f(z)$ can be written in the form of $w = f(z) = u(x, y) + iv(x, y)$.

Function of Complex numbers

Example 1

Write down $f(z) = z^2$ in the form of $f(z) = u(x, y) + iv(x, y)$.

Solution to Example 1

$$f(z) = z^2 = (x + iy)^2 \quad \text{since } z = x + iy$$

$$f(z) = (x + iy)(x + iy) = x^2 + ixy + ixy + i^2y^2$$

$$f(z) = x^2 + 2ixy - y^2$$

$$f(z) = x^2 - y^2 + 2ixy$$

$$\text{Hence } u(x, y) = x^2 - y^2 \text{ and } v(x, y) = 2xy$$

Example 2

Express $f(z) = z^2 + 3z + 4$ in the form of $f(z) = u(x, y) + iv(x, y)$

Solution to example 2

let $z = x + iy$

$$\begin{aligned}f(z) &= (x + iy)^2 + 3(x + iy) + 4 \\&= x^2 + 2ixy + i^2y^2 + 3x + 3iy + 4 \\&= x^2 + 2ixy - y^2 + 3x + 3iy + 4 \\&= x^2 - y^2 + 3x + 4 + i2xy + 3iy\end{aligned}$$

$$f(x + iy) = (x^2 - y^2 + 3x + 4) + i(2xy + 3y)$$

where $u(x, y) = x^2 - y^2 + 3x + 4$, $v(x, y) = 2xy + 3y$

Function of Complex numbers

Example 3

Write down $f(z) = z^4$ in the form of $f(z) = u(x, y) + iv(x, y)$.

Solution to example 3

Let $z = x + iy$

$f(z) = f(x + iy) = (x + iy)^4$, Using the Binomial expansion, we get

$$\begin{aligned} f(x + iy) &= (x + iy)^4 \\ &= \binom{4}{0}x^4(iy)^0 + \binom{4}{1}x^3(iy)^1 + \binom{4}{2}x^2(iy)^2 + \binom{4}{3}x(iy)^3 + \binom{4}{4}x^0(iy)^4 \\ &= x^4 + 4x^3iy + 6x^2i^2y^2 + 4xi^3y^3 + i^4y^4 \\ &= x^4 - 6x^2y^2 + y^4 + i(4x^3y - 4xy^3) \\ f(x + iy) &= (x^4 - 6x^2y^2 + y^4) + i(4x^3y - 4xy^3) \end{aligned}$$

where $u(x, y) = x^4 - 6x^2y^2 + y^4$, $v(x, y) = 4x^3y - 4xy^3$

Function of Complex numbers CONT'D

Example 4

Express $f(z) = \bar{z} \operatorname{Re} z + z^2 + \operatorname{Im} z$ in the form of $f(z) = u(x, y) + iv(x, y)$.

Solution to example 4

We use elementary properties of $\mathbb{C}$ as follows

$$\begin{aligned} f(x + iy) &= (x - iy)x + (x + iy)^2 + y \\ &= x^2 - ixy + x^2 + 2ixy - y + y \\ &= 2x^2 + 2ixy - ixy - y^2 + y \end{aligned}$$

$$\therefore f(x + iy) = (2x^2 - y^2 + y) + ixy$$

- It may also be convenient to express the complex function in polar coordinates.

$$f(r, \theta) = u + iv = u(r, \theta) + iv(r, \theta)$$

Function of Complex numbers CONT'D

Example 5

Express $f(z) = z^5 + 4z^2 - 6$ in the polar coordinate form of $u(r, \theta) + iv(r, \theta)$.

Solution to Example 5

$$x = r \cos \theta, \quad y = r \sin \theta$$

$$\begin{aligned} f(r, \theta) &= (r \cos \theta + i \sin \theta)^5 + 4(r \cos \theta + r \sin \theta)^2 - 6 \\ &= r^5(\cos 5\theta + i \sin 5\theta) + 4(r^2 \cos 2\theta + ir^2 \sin 2\theta) - 6 \\ &= r^5 \cos 5\theta + ir^5 \sin 5\theta + 4r^2 \cos 2\theta + 4ir^2 \sin 2\theta - 6 \\ &= (r^5 \cos 5\theta + 4r^2 \cos 2\theta - 6) + i(r^5 \sin 5\theta + 4r^2 \sin 2\theta). \end{aligned}$$

Limits of Functions of complex variables

Definition 2

Let $w = f(z)$ be a function defined in a domain D except perhaps at, a point $z_0 \in D$. We say that the limit of the function $f(z)$ as z approaches z_0 equals w_0 , and we write

$$\lim_{z \rightarrow z_0} f(z) = w_0 \quad \text{or} \quad f(z) \rightarrow w_0 \text{ as } z \rightarrow z_0$$

Similarly, we can write, for every $\varepsilon > 0$, there exists a number $\delta > 0$ such that $|f(z) - w_0| < \varepsilon$ whenever $0 < |z - z_0| < \delta$.

Limits of Functions of complex variables

Theorem 2

If the limit of a function exists, then it is unique.

Proof.

Suppose that $\lim_{z \rightarrow z_0} f(z) = W_1$ and $\lim_{z \rightarrow z_0} f(z) = W_2$. Then given $\varepsilon > 0$, there exists $\delta_1 > 0, \delta_2 > 0$ such that

$$|f(z) - w_1| < \frac{\varepsilon}{2} \text{ when } 0 < |z - z_0| < \delta_1$$

$$|f(z) - w_2| < \frac{\varepsilon}{2} \text{ when } 0 < |z - z_0| < \delta_2$$

Hence for $\delta = \min(\delta_1, \delta_2)$, we get

$$\begin{aligned} |w_2 - w_1| &= |(f(z) - w_1) - (f(z) - w_2)| \\ &\leq |f(z) - w_1| + |f(z) - w_2| \\ &< \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon \end{aligned}$$

when $0 < |z - z_0| < \delta$

Limits of Functions of complex variables

Example 6

Show that $\lim_{z \rightarrow 2i} \frac{z^2 + 4}{z - 2i} = 4i$

Solution

$f(z) = \frac{(z^2+4)}{(z-2i)}$ is defined everywhere except at $z = 2i$. i.e

For $z = 2i$, the numerator $z^2 + 4$ becomes:

$$z^2 + 4 = (2i)^2 + 4 = -4 + 4 = 0$$

And the denominator $z - 2i$ becomes:

$$z - 2i = 2i - 2i = 0$$

Both the numerator and denominator are zero at $z = 2i$, then we have a $\frac{0}{0}$ indeterminate form.

$$\begin{aligned}\lim_{z \rightarrow 2i} \frac{z^2 + 4}{z - 2i} &= \lim_{z \rightarrow 2i} \frac{(z - 2i)(z + 2i)}{z - 2i} \\ &= \lim_{z \rightarrow 2i} z + 2i \\ &= 2i + 2i \\ &= 4i\end{aligned}$$

Limits of Functions of complex variables

Theorem 3

Let $w = f(z) = u(z) + iv(z)$ be defined in a domain D except possibly at the point z_0 of D . Then

$$\lim_{z \rightarrow z_0} f(z) = w_0 = u_0 + iv_0 \text{ if and only}$$

$$\lim_{z \rightarrow z_0} u(z) = u_0 \quad \text{and} \quad \lim_{z \rightarrow z_0} v(z) = v_0$$

Proof

The proof is left for the reader as an exercise

Example 7

Evaluate $\lim_{z \rightarrow (1+i)} (z^2 - 2z + 1)$

Solution

$$\begin{aligned}f(z) &= z^2 - 2z + 1 = (x + iy)^2 - 2(x + iy) + 1 \\&= x^2 + 2ixy + i^2y^2 - 2x - 2iy + 1 \\&= (x^2 - y^2 - 2x + 1) + i(2xy - 2y) \\ \lim_{(x,y) \rightarrow (1,1)} u(x,y) &= \lim_{(x,y) \rightarrow (1,1)} (x^2 - y^2 - 2x + 1) \\&= 1 - 1 - 2 + 1 = -1 \\ \lim_{(x,y) \rightarrow (1,1)} v(x,y) &= \lim_{(x,y) \rightarrow (1,1)} (2xy - 2y) \\&= 2(1)(1) - 2(1) \\&= 0\end{aligned}$$

Therefore $\lim_{z \rightarrow (1+i)} f(z) = -1$

Example 8

Find $\lim_{z \rightarrow (2+i)} (z^2 - 4z + 2 + 5i)$

Solution

$$\begin{aligned}f(z) &= f(x + iy) = (x + iy)^2 - 4(x + iy) + 2 + 5i \\&= x^2 + 2ixy + i^2y^2 - 4x - 4iy + 2 + 5i \\&= x^2 + 2ixy - y^2 - 4x - 4iy + 2 + 5i \\&= (x^2 - y^2 - 4x + 2) + i(2xy - 4y + 5)\end{aligned}$$

$$\begin{aligned}\lim_{(x,y) \rightarrow (2,1)} u(x,y) &= \lim_{(x,y) \rightarrow (2,1)} x^2 - y^2 - 4x + 2 \\&= 2^2 - 1^2 - 4(2) + 2 \\&= 4 - 1 - 8 + 2 = 3 - 8 + 2 \\&= -5 + 2 = -3\end{aligned}$$

$$\begin{aligned}\lim_{(x,y) \rightarrow (2,1)} v(x,y) &= \lim_{(x,y) \rightarrow (2,1)} (2xy - 4y + 5) = 2(2)(1) - 4(1) + 5 \\&= 4 - 4 + 5 = 5\end{aligned}$$

$$\therefore \lim_{z \rightarrow (2+i)} f(z) = -3 + 5i$$

Limits of Functions of complex variables

Example 9

Find $\lim_{z \rightarrow (1-i)} z^2 + 5z + 2i$

Solution

$$\begin{aligned}\lim_{z \rightarrow (1-i)} z^2 + 5z + 2i &= (1-i)^2 + 5(1-i) + 2i \\ &= 1 - 2i + i^2 + 5 - 5i + 2i \\ &= 1 - 1 + 5 - 5i \\ &= 5 - 5i\end{aligned}$$

Example 10

Find the value of $\lim_{z \rightarrow (1+i)} \left(\frac{z^2 - 2i}{z^2 - 2z + 2} \right)$

Solution

Define $P(z) = z^2 - 2i$ and $Q(z) = z^2 - 2z + 2$.

Factoring out $P(z)$ and $Q(z)$ we get $P(z) = (z - 1 - i)(z + 1 + i)$,
 $Q(z) = (z - 1 + i)(z - 1 - i)$

$$\lim_{z \rightarrow 1+i} \frac{z^2 - 2i}{z^2 - 2z + 2} = \lim_{z \rightarrow (1+i)} \frac{(z - 1 + i)(z + 1 + i)}{(z - 1 + i)(z + 1 - i)} = \lim_{z \rightarrow (1+i)} \frac{z + 1 + i}{z - 1 + i}$$

$$= \frac{1 + i + 1 + i}{1 + i - 1 + i} = \frac{2 + 2i}{2i} = \frac{2(1 + i)}{2i} \quad \text{But } 2 = (1 + i)(1 - i)$$

$$\frac{2(1 + i)}{2i} = \frac{(1 + i)(1 - i)(1 + i)}{2i} \quad \text{Since } (1 + i)(1 - i) = 2i$$

$$\text{Then } \frac{2i(1 - i)}{2i} = 1 - i$$

Limits of Functions of complex variables

Theorem 4

Let $\lim_{z \rightarrow z_0} f(z) = A$ and $\lim_{z \rightarrow z_0} g(z) = B$, then

(a) $\lim_{z \rightarrow z_0} f(z) \pm g(z) = A \pm B$

(b) $\lim_{z \rightarrow z_0} f(z)g(z) = AB$

(c) $\lim_{z \rightarrow z_0} \frac{f(z)}{g(z)} = \frac{A}{B}$, for $B \neq 0$

(d) $\lim_{z \rightarrow z_0} kf(z) = kA$, for k a constant

CONTINUOUS FUNCTIONS OF A COMPLEX VARIABLE

Definition

The function $w = f(z)$ defined on a domain D is said to be continuous at the point z_0 in D if $\lim_{z \rightarrow z_0} f(z) = f(z_0)$.

• This definition actually asserts that $w = f(z)$ is continuous at z_0 in D if

- (i) $f(z_0)$ is defined
- (ii) $\lim_{z \rightarrow z_0} f(z)$ exists
- (iii) $\lim_{z \rightarrow z_0} f(z) = f(z_0)$

CONTINUITY

Alternative Definition

Alternatively, a function $f(z)$ is continuous at a point z_0 in D if and only if for every $\varepsilon > 0$, there is a $\delta > 0$ such that

$$|f(z) - f(z_0)| < \varepsilon$$

for all points z interior to D satisfying

$$|z - z_0| < \delta$$

Theorem 5

If the function $f(z)$ and $g(z)$ are defined in a domain D and are continuous at the point z_0 in D , then

(a) $f(z) + g(z)$

(b) $f(z) - g(z)$

(c) $f(z)g(z)$

(d) $kf(z)$

(e) $f(z)/g(z), \quad g(z_0) \neq 0$

are continuous at the point z_0

Definition

A polynomial function $P(z) = a_n z^n + \dots + a_1 z + a_0$ is continuous at any point z_0 because $\lim_{z \rightarrow z_0} P(z) = P(z_0)$.

Definition

A rational function $f(z) = \frac{P(z)}{Q(z)}$ where $P(z)$ and $Q(z)$ are polynomials is continuous at any point z_0 provided that $Q(z_0) \neq 0$.

Theorem 5

If $f(z)$ is continuous and has the value w_0 at the point z_0 and if $F(w)$ is continuous at w_0 , then $F(f(z))$ is continuous at z_0 .

Proof.

Proof is left for the you as an exercise



CONTINUITY

Theorem 6

Let $f(z)$ be a continuous function on a compact set R , then $f(z)$ is uniformly continuous on R

Example 11

Let R be a set of all points z such that $0 < |z| \leq 1$. show $f(z) = \frac{1}{z}$ is continuous on R .

Solution

Using the definition of continuity, $f(z) = \frac{1}{z}$ is defined on all R . Also $\lim_{z \rightarrow z_0} \frac{1}{z} = \frac{1}{z_0}$ for all $z_0 \in R$. Hence $f(z)$ is continuous on R .

Example 12

Let R be a set of all points z such that $0 < |z| \leq 1$. Show that $f(z) = z^2$ is uniformly continuous on R .

solution

Let z_1 and z_2 be in R , then we have

$$\begin{aligned} |f(z_1) - f(z_2)| &= |z_1^2 - z_2^2| = |(z_1 - z_2)(z_1 + z_2)| \\ &= |z_1 + z_2| |z_1 - z_2| \leq 2|z_1 - z_2| \end{aligned}$$

since $0 < |z| \leq 1$. Taking $\delta = \frac{\epsilon}{2}$, then for any two points $z_1, z_2 \in R$ such that $|z_1 - z_2| < \delta$, we have

$$|f(z_1) - f(z_2)| < \epsilon.$$

Thus $f(z) = z^2$ is uniformly continuous on R .

THE END